Figure 12.4 Get movie by ID request, showing the test results from within Postman

The Postman application can add tests from within the interface, and it also provides the option for JavaScript tests. As shown in figure 12.4, the tests verify the headers, status, and body of the response. As with the previous Supertest example, we can translate this test into a contract test, as shown in listing 12.5. The `uponReceiving` method sets the state of the data required; a movie with ID 1 will need to be in the database in this example. The `withRequest` method sets the request method and path to get a movie by ID. The `willRespondWith` method defines the expected headers, status code, and response body. The `body` contains matches of type integer and string to verify values against their type only. After the provider expectations have been set up, then the request to get the movie is executed with the mock provider URL and ID 1. The `expect` statement verifies that the response object isn't null.

Listing 12.5 Pact get movie by ID

```
test('it should return a movie object', async () => {
  provider
    .uponReceiving('get movie by id 1')
    .withRequest({
      method: 'GET',
      path: '/movie/1',
    })
    .willRespondWith({
      status: 200,
      headers: { 'Content-Type': 'application/json' },
      body: {
        id: integer(1),
        name: string("Barbie"),
        year: integer(2023)
      },
    });

  await provider.executeTest(async mockProvider => {
    const movie = await fetchSingleMovie(mockProvider.url, 1);
    expect(movie).not.toBeNull();
  });
});
```

Once the test has been run, the contract is generated with the interaction setup in figure 12.5. The contract shown in listing 12.6 can now be published to Pact Broker and be shared with the provider. The provider can then set up the data required and run against the local instance of the service to verify the contract. The benefit of converting this test is that the tests can be run in isolation. Another benefit is being able to set up the data